EECS 245, Fall 2026

LEC 12 Rank and Column Space

→ Read Ch. 5.2-5.4

HAPPY BIRTHDAY

My freshman year transcript

Fall 2016

Class	Title	Un.	Gr.
CHEM 1A	General Chemistry	3	B-
CHEM 1AL	General Chemistry Laboratory	1	C+
COMPSCI 61A	The Structure and Interpretation of Computer Programs	4	B+
COMPSCI 70	Discrete Mathematics and Probability Theory	4	A
COMPSCI 195	Social Implications of Computer Technology	1	P
MATH 1A	Calculus	4	A+

Spring 2017

Class	Title	Un.	Gr.
COMPSCI 61B	Data Structures	4	B+
COMPSCI 97	Field Study	1	P
COMPSCI 197	Field Study	1	P
ELENG 16A	Designing Information Devices and Systems I	4	B-
MATH 110	Linear Algebra	4	C
MATH 128A	Numerical Analysis	4	B+

Kind of like EECS 245

Math 217

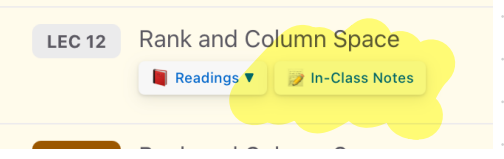
Agenda

- Recap: Matrices and the transpose operation 5.1-5.2
- The column space and rank of a matrix 5.3
- If time: basics of the null space of a matrix 5.4

Announcements

solutions posted too!

- Midterm 1 scores posted: regrades due Thursday
  - Come to OH to discuss performance! Zoom available.
- HW 5 due Sunday
- Follow along live:



Last time: simplify

$A_{n \times d}$ $\vec{x}$ vector in $\mathbb{R}^d$ $d \times 1$

$$\|A\vec{x}\|^2 = (A\vec{x}) \cdot (A\vec{x})$$
$$= (A\vec{x})^T (A\vec{x})$$
$$= \vec{x}^T A^T A \vec{x}$$

$A\vec{x}$: vector in $\mathbb{R}^n$ $n \times 1$

$(AB)^T = B^T A^T$

$A^T A$ contains the dot products of all pairs of A's columns!

A^TA : dot products of all pairs of A's columns

review!

A $n \times d$

A^T A

$d \times d$

square and equal to its transpose!

A^TA is a symmetric matrix!

A^T $n \times n$ A $n \times d$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

not symmetric!

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

symmetric!

$$(A^TA)^T = A^T(A^T)^T$$
$$= A^TA \checkmark$$

$$(\square\triangle)^T = \triangle^T\square^T$$

$A^TA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ — dot products of all pairs of A's cols

What do I know about A?

- A has 2 columns (# rows unknown)
- A's first and second columns are orthogonal!
- A's columns are unit vectors!

(col 1) · (col 1) = 1 ⇒ col 1 has length of 1!

Aside: transpose

$$X = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 1 & -9 \end{bmatrix}_{2\times 3}$$

$$X^T = \begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 7 & -9 \end{bmatrix}_{3\times 2}$$

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}_{4\times 1} \qquad \vec{v} = \begin{bmatrix} -9 \\ 1 \\ 4 \\ 7 \end{bmatrix}_{4\times 1}$$

$$\vec{u}\cdot\vec{v} = \vec{u}^T\vec{v}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}\begin{bmatrix} -9 \\ 1 \\ 4 \\ 7 \end{bmatrix}$$

$$= 1(-9) + 2(1) + \cdots$$

$$= \vec{v}^T\vec{u} = \vec{v}\cdot\vec{u}$$

$\vec{u}^T\vec{v}$ (1×4 4×1) : output is 1×1, i.e. a scalar!

Column space (5.3)

$A\vec{x}$ is just a linear combination of A's cols!

$$A = \begin{bmatrix} 5 & 3 & 2 \\ 0 & -1 & 1 \\ 3 & 4 & -1 \\ 6 & 2 & 4 \\ 1 & 0 & 1 \end{bmatrix}_{5\times 3} \qquad \vec{x} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

$$A\vec{x} = (2)\begin{bmatrix} 5 \\ 0 \\ 3 \\ 6 \\ 1 \end{bmatrix} + (0)\begin{bmatrix} 3 \\ -1 \\ 4 \\ 2 \\ 0 \end{bmatrix} + (-1)\begin{bmatrix} 2 \\ 1 \\ -1 \\ 4 \\ 1 \end{bmatrix}$$

Column space of a matrix A,

$\text{colsp}(A)$ is

- the span of A's columns
- i.e. set of all possible linear combinations of A's columns
- i.e. set of all possible outputs of $A\vec{x}$, for all possible $\vec{x}$'s

$$A_{n \times d} = \begin{bmatrix} | & | & & | \\ \vec{a}^{(1)} & \vec{a}^{(2)} & \cdots & \vec{a}^{(d)} \\ | & | & & | \end{bmatrix}$$

$\vec{a}^{(i)}$: col i of A

$\vec{a}^{(i)} \in \mathbb{R}^n$

$$\text{colsp}(A) = \text{span}\left(\left\{ \vec{a}^{(1)}, \vec{a}^{(2)}, \ldots, \vec{a}^{(d)} \right\}\right)$$

subspace of $\mathbb{R}^n$

$$A = \begin{bmatrix} 5 & 3 & 2 \\ 0 & -1 & 1 \\ 3 & 4 & -1 \\ 6 & 2 & 4 \\ 1 & 0 & 1 \end{bmatrix}$$

col 1 col 2 col 3

col 1 = col 2 + col 3, so one column is redundant!

① $\dim(\text{colsp}(A)) = 2$

colsp(A) is a 2-dimensional subspace of $\mathbb{R}^5$.

$$\text{colsp}(A) = \text{span}\left(\left\{\begin{bmatrix} 3 \\ -1 \\ 4 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \\ 4 \\ 1 \end{bmatrix}\right\}\right)$$

one of infinitely many bases

$$\text{rank}(A) = \dim(\text{colsp}(A))$$

$$= \#\text{ linearly independent columns}$$

don't forget this!
most important number
associated with a matrix.

Activity: Think of a 3×4 matrix with

(1) rank 1

$$\begin{bmatrix} 1 & 2 & 7 & 4 \\ 2 & 4 & 14 & 8 \\ 3 & 6 & 21 & 12 \end{bmatrix}$$

(2) rank 2

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 5 \\ 1 & 0 & \pi & 412 \end{bmatrix}$$

(3) rank 3

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 14 \\ 0 & 0 & 1 & 21 \end{bmatrix}$$

(4) rank 4

not possible!

can't have 4 LI vectors in $\mathbb{R}^3$.

$$A = \begin{bmatrix} 5 & 3 & 2 \\ 0 & -1 & 1 \\ 3 & 4 & -1 \\ 6 & 2 & 4 \\ 1 & 0 & 1 \end{bmatrix} \qquad A^T = \begin{bmatrix} 5 & 0 & 3 & 6 & 1 \\ 3 & -1 & 4 & 2 & 0 \\ 2 & 1 & -1 & 4 & 1 \end{bmatrix}$$

$\vec{a}_2$ $\vec{a}_5$

$\vec{a}_1, \vec{a}_3, \vec{a}_4$ can be written as LCs of $\vec{a}_2, \vec{a}_5$!

$\dim(\text{colsp}(A)) = 2$

$\dim(\text{colsp}(A^T)) = 2$

row space of A, set of all lin combs of A's rows

$\Rightarrow$ all 5 of A's rows satisfy $x - y - z = 0$

plane in $\mathbb{R}^3$!

Fact: For any matrix A,

$$\dim(\text{colsp}(A)) = \text{rank}(A)$$

$\dim(\text{colsp}(A))$: # linearly independent cols

is equal to:

$$\dim(\text{colsp}(A^T)) = \text{rank}(A^T)$$

$\dim(\text{colsp}(A^T))$: # linearly independent rows

"tall"

~ columns could all be LI,
but rows cannot!

$0 \leq \text{rank}(A) \leq \min(n, d)$